Syed Talha Bukhari

ML Software Engineer, Pointivo, Inc.

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Research Interests

Machine Learning, Computer Vision, Robotics.

Education

2018 *to* 2020 **M.S. Electrical Engineering**, *CGPA: 3.96/4.00*,
LUMS School of Science and Engineering (SSE), Lahore.

- Graduated in Dean's Honor List.
- Thesis/Research Focus: Deep Learning for 3D mpMRI Segmentation.
- Coursework includes: Convex Optimization, Stochastic Systems, Information Theory & Machine Learning, Computer Vision Fundamentals, Deep Learning, Advanced Digital Signal Processing, Robot Motion Planning.

2014 *to* 2018 **B.S. Mechatronics & Control Engineering**, *CGPA: 3.94/4.00*,
University of Engineering and Technology (UET), Lahore.

- Graduated with highest distinction.
- Coursework includes: Robotics, Control Systems, Digital Image Processing, Digital Signal Processing, Signals & Systems, Intelligent Systems, Embedded Systems, Numerical Methods, Hydraulics & Pneumatics.

Work Experience

8/2021 *to* **Present** **Machine Learning Software Engineer**, Pointivo, Inc., USA.

- Working, as part of the Cognitive Services team, to deploy Deep Learning-based solutions to solve tasks in physical asset inspection.

7/2020 *to* 9/2021 **Research Associate**, *Department of Electrical Engineering*,
LUMS School of Science and Engineering (SSE), Lahore.

- Led the development of Deep Learning pipelines for medical image segmentation.
- Co-supervised and guided M.S. and Ph.D. students working in the group.
- Director/Supervisor: Dr. Hassan Mohy-ud-Din

1/2019 *to* 6/2020 **Graduate Research Assistant**, *Department of Electrical Engineering*,
LUMS School of Science and Engineering (SSE), Lahore.

- Led the development of Deep Learning pipelines for medical image segmentation.
- Part-time employment, in parallel with graduate studies.
- Director/Supervisor: Dr. Hassan Mohy-ud-Din

Spring 2020 **Graduate Teaching Assistant**, *EE516: Foundations of Deep Learning*,
LUMS School of Science and Engineering (SSE), Lahore.

- Designed coursework and graded components with the instructor, in the inaugural offering of the graduate course.
- Focused on imparting fundamental skills relevant to implementation of Deep Learning in PyTorch.

07 *to* 08/2017 **Intern**, *Centre of Excellence in Solid State Physics*,
University of the Punjab, Lahore.

- Involved in device fabrication and characterization process at micro level.
- Supervisor: Prof. Dr. Shahzad Naseem

Relevant Technical Skills

Languages Python, MATLAB, C++, Julia (basics)

Frameworks PyTorch, TensorFlow (1.x & 2.x), pandas, OpenCV, JAX (basics)

Embedded Proteus (ISIS & ARES), Arduino, MPLab (basics)

Robotics V-REP (with MATLAB), ROS (basics)

Mechanical Design SolidWorks, AutoCAD

Other LaTeX, Git, Docker, Bash scripting (basics)

Talks/Workshops

11/2020 **Speaker**, *Seminar on Deep Learning*, University of Engineering & Technology, Lahore.

- A one-day seminar covering fundamental concepts in Machine Learning and Deep Learning, targeted at senior undergraduate students.

12/2019 **Co-Instructor**, *Short Course on Machine Learning and Deep Learning*, 2nd Annual Workshop on Big Data and Cloud Computing (BigC 2019).

- A one-day workshop/seminar covering fundamental concepts in Machine Learning and Deep Learning, and implementation of Neural Networks in Python with TensorFlow.

Awards & Honors

2020 LUMS Dean's Honor Roll for distinction in graduate studies.

2018–19 & 2019–20 LUMS Merit Scholarship Award for graduate studies (complete coverage of tuition for an academic year, ×2).

2018 UET Gold Medal for top rank in undergraduate studies.

2014–18 UET Dean's Honor Roll (×8) for distinction in undergraduate studies.

2018 1st Position Senior Year Project (awarded by department): Automatic Retail Checkout via Deep Learning.

2017–18 Senior Year Project approved for funding by National Grassroots ICT Research Initiative.

2014 College Best Student of the Year (12th grade).

2008 International Kangaroo Mathematics Contest: 27th position in the country (3rd in school) out of 10,556 participants at the *Benjamins* level.

Publications

Jabbar, Sana ; Bukhari, Syed T. ; Mohy-ud-Din, Hassan: On the Generalizability of Convolutional Neural Networks for Accurate Quantification of Cardiac Function on 3D MRI Scans: A Multi-vendor, Multi-center Study. In: *Physics in Medicine & Biology* (2021, **under review**)

Bukhari, Syed T. ; Din, Hassan M.: *E_1D_3 U-Net for Brain Tumor Segmentation: Submission to the RSNA-ASNR-MICCAI BraTS 2021 Challenge.* https://arxiv.org/abs/2110.02519. Version: 2021 (**accepted talk at MICCAI BrainLes 2021, under review at RSNA 2021)**

Jabbar, Sana ; Bukhari, Syed T. ; Mohy-ud-Din, Hassan: Multi-view SA-LA Net: A Framework for Simultaneous Segmentation of Right Ventricle on Multi-View Cardiac MR Images. In: *International Workshop on Statistical Atlases and Computational Models of the Heart (STACOM 2021)* Springer, 2021

Bukhari, Syed T. ; Mohy-ud-Din, Hassan: A Systematic Evaluation of Learning Rate Policies in Training CNNs for Brain Tumor Segmentation. In: *Physics in Medicine & Biology* [IF:3.61] (2021). https://iopscience.iop.org/article/10.1088/1361-6560/abe3d3

Bukhari, Syed T. ; Amin, Abdul W. ; Naveed, Muhammad A. ; Abbas, Muhammad R.: *ARC: A Vision-based Automatic Retail Checkout System.* https://arxiv.org/abs/2104.02832. Version: 2021